

MATH 151A MIDTERM 2 FORMULA SHEET

1. LAGRANGE INTERPOLATION

Lagrange Basis Polynomials

$$l_j(x) = \prod_{k=0, k \neq j}^n \frac{x - x_k}{x_j - x_k}$$

Property: $l_j(x_i) = \delta_{ij}$ (equals 1 if $i=j$, else 0)

Interpolating Polynomial

$$p_n(x) = \sum_{j=0}^n f(x_j) \cdot l_j(x)$$

Ψ Notation

$$\Psi_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$$

$$l_j(x) = \frac{\Psi_n(x)}{(x - x_j)\Psi'_n(x_j)}$$

Error Formula

$$f(x) - p_n(x) = \Psi_n(x) \cdot f[x_0, \dots, x_n, x] = \frac{\Psi_n(x)}{(n+1)!} f^{(n+1)}(\xi)$$

2. DIVIDED DIFFERENCES

Recursive Definition

- **Zeroth:** $f[x_i] = f(x_i)$
- **First:** $f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$
- **Second:** $f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$
- **General:** $f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$

Newton's Divided Difference Form

$$p_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

Expanded: $p_n(x) = f[x_0] + f[x_0, x_1] + f[x_0, x_1, x_2](x - x_1) + \dots$

Connection to Derivatives

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}$$

When nodes coincide:

- $f[x_i, x_i] = f'(x_i)$
 - $f[x_i, x_i, x_i] = \frac{f''(x_i)}{2!}$
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3. HERMITE INTERPOLATION

Cubic Hermite (4 conditions: $f(a)$, $f'(a)$, $f(b)$, $f'(b)$)

$$H_2(x) = f(a) + (x - a)f'(a) + (x - a)^2 f[a, a, b] + (x - a)^2 (x - b) f[a, a, b, b]$$

Where:

$$f[a, b] = \frac{f(b) - f(a)}{b - a}$$

$$f[a, a, b] = \frac{f[a, b] - f'(a)}{b - a}$$

$$f[a, a, b, b] = \frac{f'(b) - 2f[a, b] + f'(a)}{(b - a)^2}$$

Error Formula

$$f(x) - H_2(x) = \frac{(x - a)^2 (x - b)^2}{24} f^{(4)}(\xi)$$

Maximum error:

$$\max_{x \in [a, b]} |f(x) - H_2(x)| \leq \frac{(b-a)^4}{384} \max |f^{(4)}(t)|$$

4. SPLINES

Definition

Spline of order m:

1. Degree $< m$ on each subinterval $[x_{i-1}, x_i]$
2. C^{m-2} continuous on $[a, b]$

Cubic Spline (order 4):

- Degree 3 polynomials on each piece
- C^2 continuous (s, s', s'' all continuous)

Degrees of Freedom

For n subintervals:

- **Unknowns:** $4n$ coefficients
- **Constraints:** $(n+1) + 3(n-1) = 4n-2$
- **Need:** 2 boundary conditions

Common Boundary Conditions

1. **Clamped:** $s'(a) = f'(a), s'(b) = f'(b)$
2. **Natural:** $s''(a) = 0, s''(b) = 0$
3. **Not-a-knot:** s''' continuous at x_1 and x_{n-1}
4. **Periodic:** $s'(a) = s'(b), s''(a) = s''(b)$

Key Property (Minimizes Curvature)

$$\int_a^b |s_c''(x)|^2 dx \leq \int_a^b |g''(x)|^2 dx$$

for any other interpolant g

Error Bound (Clamped Spline)

$$\max |f^{(j)}(x) - s_c^{(j)}(x)| \leq c_j h^{4-j} \max |f^{(4)}(x)|$$

where $h = \max_i (x_i - x_{i-1})$ and:

- $c_0 = \frac{5}{384}$
 - $c_1 = \frac{1}{24}$
 - $c_2 = \frac{3}{8}$
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5. NUMERICAL DIFFERENTIATION

Basic Formula

$$f'(x) \approx p'_n(x) = \sum_{j=0}^n f(x_j) l'_j(x)$$

Error

$$f'(x) - p'_n(x) = \Psi'_n(x) \frac{f^{(n+1)}(\xi_1)}{(n+1)!} + \Psi_n(x) \frac{f^{(n+2)}(\xi_2)}{(n+2)!}$$

For evenly spaced nodes (spacing h):

$$\text{Error} = \begin{cases} O(h^n) & \text{if } \Psi'_n(x) \neq 0 \\ O(h^{n+1}) & \text{if } \Psi'_n(x) = 0 \end{cases}$$

Three Key Formulas

1. Forward Difference (2-point)

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}$$

- Error: $\frac{h}{2} f''(\xi)$
- Order: $O(h)$

2. Centered Difference (2-point) ★ BEST

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

- Error: $-\frac{h^2}{6} f'''(\xi)$
- Order: $O(h^2)$

3. Three-Point Endpoint

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_1) - f(x_2)}{2h}$$

where $x_1 = x_0 + h, x_2 = x_0 + 2h$

- Error: $\frac{h^2}{3} f'''(\xi)$
- Order: $O(h^2)$

Round-off Error

$$E_{total} = Ch^p + \frac{\epsilon}{h}$$

- As $h \rightarrow 0$: truncation $\downarrow$ but round-off $\uparrow$
 - Optimal: $h_{opt} \approx \left(\frac{\epsilon}{C}\right)^{1/(p+1)}$
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6. NUMERICAL INTEGRATION

A. Trapezoidal Rule

Simple Trapezoidal:

$$\int_a^b f(x)dx \approx \frac{b-a}{2}[f(a) + f(b)]$$

- Error: $-\frac{(b-a)^3}{12} f''(\eta)$

Composite Trapezoidal (n subintervals):

$$I_n(f) = h \left[\frac{f(a) + f(b)}{2} + \sum_{i=1}^{n-1} f(x_i) \right]$$

where $h = \frac{b-a}{n}$

- Error: $-\frac{(b-a)^3}{12n^2} f''(\eta) = -\frac{(b-a)h^2}{12} f''(\eta)$
- Order: $O(h^2)$

B. Simpson's Rule

Simple Simpson's:

$$\int_a^b f(x)dx \approx \frac{h}{3}[f(a) + 4f(c) + f(b)]$$

where $c = \frac{a+b}{2}$, $h = \frac{b-a}{2}$

- Error: $-\frac{h^5}{90}f^{(4)}(\eta) = -\frac{(b-a)^5}{2880}f^{(4)}(\eta)$

Composite Simpson's (n even subintervals):

$$S_n(f) = \frac{h}{3} \left[f(a) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(b) \right]$$

where $h = \frac{b-a}{n}$

- Error: $-\frac{(b-a)^5}{180n^4}f^{(4)}(\eta)$
- Order: $O(h^4)$ ★ EXCELLENT

Computing Integration Weights

For nodes $x_0, \dots, x_n$:

$$w_j = \int_a^b l_j(x)dx$$

Then:

$$\int_a^b f(x)dx \approx \sum_{j=0}^n w_j f(x_j)$$

Integral Mean Value Theorem

If g(x) doesn't change sign:

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$$

If g(x) changes sign: Cannot use MVT directly!

QUICK REFERENCE TABLE

Method	Nodes	Error Order	Best Use
Lagrange n	n+1	$O(h^{n+1})$	Simple interpolation
Hermite Cubic	2 (with derivatives)	$O(h^4)$	When derivatives known
Cubic Spline	n+1	$O(h^4)$	Smooth curves, avoid oscillation
Forward Diff	2	$O(h)$	Avoid if possible
Centered Diff	2	$O(h^2)$	★ Best for derivatives
3-Point Endpoint	3	$O(h^2)$	When centered not available
Trapezoidal	2	$O(h^2)$	Simple integration
Simpson's	3	$O(h^4)$	★ Best integration

KEY FACTS TO REMEMBER

Vandermonde Matrix

Solve $Xa = y$ where $X_{i,j} = x_i^j$ to find polynomial coefficients

Why Splines Beat High-Degree Polynomials

- High-degree $\rightarrow$ Runge phenomenon (wild oscillation)
- Cubic splines minimize $\int |s''|^2 \rightarrow$ smoothest
- Better approximation with lower degree

Why Centered Schemes Are Better

- Symmetric nodes $\rightarrow \Psi'(x) = 0$ at center point
- Get extra order of accuracy for free
- $O(h^2)$ vs $O(h)$ for forward difference

Round-off vs Truncation Trade-off

- Small h: good truncation error, BAD round-off
- Large h: BAD truncation error, good round-off
- Don't make h too small!

When Integral MVT Applies

- Need $g(x)$ to NOT change sign
- For Trapezoidal: $(x-a)(x-b) < 0$ everywhere $\rightarrow$ OK ✓
- For Simpson's: $(x-a)(x-c)(x-b)$ changes sign $\rightarrow$ Need different approach

ROTE MEMORIZATION CHECKLIST

- ✓ Lagrange basis: $l_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$
- ✓ Divided difference recursion formula
- ✓ Cubic Hermite: $f[a, a, b]$ and $f[a, a, b, b]$ formulas
- ✓ Spline definition: degree $< m$, C^{m-2} continuous
- ✓ Cubic spline: $4n$ unknowns, $4n-2$ equations, need 2 BC
- ✓ Centered difference: $\frac{f(x+h) - f(x-h)}{2h}$ with $O(h^2)$ error
- ✓ Trapezoidal: $\frac{b-a}{2}[f(a) + f(b)]$ with $-\frac{(b-a)^3}{12}f''(\eta)$ error
- ✓ Simpson's: $\frac{h}{3}[f(a) + 4f(c) + f(b)]$ with $-\frac{h^5}{90}f^{(4)}(\eta)$ error